Multi-Modal Transformer for Text-Conditioned Land Cover Segmentation

Kseniia Khudiakova
Middle East Technical University
Email: e278979@metu.edu.tr

***Abstract*—This project explores text-conditioned semantic segmentation for land cover classification using transformer-based architectures. In Phase 1, a preliminary multi-modal model with CLIP vision encoder and simple feature concatenation was tested. In Phase 2, the architecture was refined to native SegFormer-B0 with FiLM conditioning. Five caption sources were evaluated against a strong vision-only baseline. While multi-modal models show competitive validation accuracy and improvements on rare classes, they have not yet surpassed the baseline in overall mIoU. The impact of caption quality remains a key factor.**

I. Introduction

Remote sensing datasets increasingly include multi-modal annotations—RGB images, pixel-level segmentation masks, and textual descriptions. While vision transformers excel at segmentation [1], they lack high-level semantic understanding present in textual captions. This project addresses the gap by developing a text-conditioned segmentation model and evaluating whether textual information improves performance over a vision-only baseline on a 7-class land cover dataset.

II. Literature Survey

Vision transformers such as SegFormer [1] achieve state-of-the-art performance in semantic segmentation using hierarchical attention, outperforming CNN-based approaches in remote sensing [4]. However, they rely solely on visual input and often struggle with underrepresented classes. Multi-modal transformers address this by integrating visual and textual information. Models like LLaVA [2] enable joint image-text reasoning via CLIP-based encoders and LLMs, while SAM [3] introduces prompt-based segmentation with textual or spatial guidance.

Despite these advances, most work focuses on general vision-language tasks, while text-conditioned segmentation in remote sensing remains underexplored, particularly regarding the impact of automatically generated captions.

This project studies how different captioning models affect segmentation quality in a multi-modal transformer framework.

III. Research Questions

- How accurately can segmentation masks be generated from RGB images combined with textual descriptions, compared to ground-truth masks?
- Which captioning model (from given dataset) produces descriptions most useful for accurate mask reconstruction?
- Can text-conditioned multi-modal models outperform a strong vision-only baseline (SegFormer-B0) on this dataset?

IV. Proposed Method

We compare two segmentation approaches, with the final multi-modal design evolving from a proof-of-concept experiment (Section V).

Baseline (Vision-Only): A SegFormer-B0 [1] processes RGB images directly to predict segmentation masks; it is trained with strong augmentations on the full dataset and serves as the reference model.

Multi-Modal (Image+Text): The architecture conditions a SegFormer-B0 decoder on textual information via FiLM (Feature-wise Linear Modulation) layers. For each input:

1) Load the RGB image, the corresponding caption, and the ground-truth mask.
2) **Vision Encoder**: The native SegFormer-B0 hierarchical encoder extracts multi-scale visual features.
3) **Text Encoder**: Captions are encoded into a fixed-size embedding with `all-MiniLM-L6-v2` (Sentence-BERT).
4) **Fusion**: The text embedding is projected and then injected into every decoder block through FiLM layers, modulating feature maps at multiple scales.
5) **Decoder**: The standard SegFormer decoder produces the final segmentation mask.

Five sources of automatically generated captions are evaluated: `text_qwen3-4b`, `hybrid_gemma3-4b`, `hybrid_qwen3-vl-8b`, `vision_gemma3-4b`, and `vision_qwen3-vl-8b`. All models are trained on the full dataset using a stratified 70/15/15 split, optimized with cross-entropy loss, and compared against the vision-only baseline via mean IoU and per-class IoU.

V. Phase 1: Proof of Concept

As an initial feasibility test, we implemented a simpler multi-modal architecture on a small data subset. The vision encoder was a frozen CLIP-ViT-L/14, the text encoder was frozen MiniLM-L6, and the two modalities were fused by element-wise addition after projection to 256 dimensions. The decoder was a randomly initialised SegFormer-B0 head. Training used only 1 000 images (10% of the full dataset) with an 80/20 split, AdamW ($lr = 5 \times 10^{-4}$), batch size 4, and 5 epochs.

TABLE I
Proof-of-concept validation results.

Model	Pixel Acc	mIoU
SegFormer-B0 (RGB-only)	0.878	0.347
Ours (Image+Text)	0.844	0.361

Qualitative Findings.

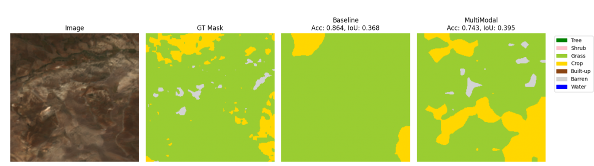

Fig. 1. Qualitative comparison between baseline and multi-modal models.

As Fig. 1 shows, the multi-modal model identified more classes, although boundaries were less precise – consistent with higher mIoU

TABLE II
BENCHMARKING RESULTS

Model	Caption Source	Val Acc	Test mIoU
Baseline (Vision-only)	—	0.910	0.611
mm-text_qwen3-4b-finetune	text_qwen3-4b (fine-tuned)	0.874	0.552
mm-hybrid_qwen3-vl-8b	hybrid_qwen3-vl-8b	0.874	0.518
mm-hybrid_gemma3-4b	hybrid_gemma3-4b	0.876	0.523
mm-vision_gemma3-4b	vision_gemma3-4b	0.869	0.545
mm-vision_qwen3-vl-8b	vision_qwen3-vl-8b	0.865	0.540

despite lower pixel accuracy. The positive signal motivated the switch to a more integrated fusion strategy (FiLM) and training on the full dataset in Phase 2.

VI. Phase 2: Benchmarking and Preliminary Results

A. Methodology Updates

Following the proof-of- feedback, the architecture was updated based on observed limitations. The CLIP vision encoder was replaced with the native SegFormer-B0 encoder. Text embeddings extracted by `all-MiniLM-L6-v2` are injected into the decoder via Feature-wise Linear Modulation (FiLM) layers. This design enables more effective cross-modal interaction while maintaining compatibility with the SegFormer architecture. All models were trained on the full dataset using a stratified 70/15/15 train/val/test split (stratified by dominant class).

Training details: image size 384×384, batch size 8–12, learning rate 1×10^{-4}, AdamW optimizer, CosineAnnealingLR scheduler, up to 50 epochs with early stopping.

(Note: the fine-tuned run means that it was terminated early by a Kaggle session timeout, and then continued using same weights).

B. Benchmarking Setup

We compare:

- **Baseline**: Vision-only SegFormer-B0 trained with augmentations on the full dataset.
- **Multi-modal variants**: FiLM-conditioned models using different caption sources (`text_qwen3-4b`, `hybrid_gemma3-4b`, `hybrid_qwen3-vl-8b`, etc.).

Evaluation metrics: mean Intersection over Union (mIoU), per-class IoU, and validation accuracy.

C. Results

Table II summarizes the main benchmarking results on the test set. All multi-modal models use the same SegFormer-B0 backbone with FiLM conditioning and `all-MiniLM-L6-v2` text encoder. Validation accuracy and test mIoU are reported.

The best multi-modal model (`mm-text_qwen3-4b-finetune`) achieves a test mIoU of 0.552, followed closely by `mm-vision_gemma3-4b` at 0.545. Both are competitive with the strong vision-only baseline (0.611), but do not surpass it. Fine-tuning the text encoder for the text-based captions yielded a noticeable boost over the default frozen setup (which had crashed during training). Several multi-modal variants demonstrate promising per-class improvements, particularly on underrepresented classes.

Per-class IoU analysis (available in the Wandb project and at Fig 2) shows that textual conditioning helps with classes such as *Crop*, and *Water*, where visual cues alone are often insufficient.

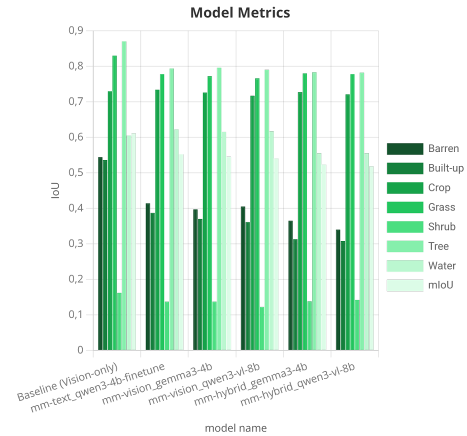

Fig. 2. Per-class IoU comparison between baseline and best multi-modal model.

D. Preliminary Discussion

Preliminary results indicate that while multi-modal models do not yet significantly outperform the carefully tuned vision-only baseline, they demonstrate promising behavior on rare classes. The quality and style of generated captions appear to be a critical factor — hybrid captions (combining vision and text) tend to perform better. Limitations include potential noise in automatically generated captions and the need for more sophisticated fusion or fine-tuning strategies.

These findings motivate further ablation studies on caption quality, FiLM placement, and text-embedding fusion techniques in Phase 3.

VII. Conclusion

Phase 1 demonstrated that even a simple additive text-conditioned fusion could improve class-wise segmentation on a small subset (+1.4 mIoU, 0.347 → 0.361), albeit with a drop in pixel accuracy.

Phase 2 scaled the approach to the full dataset and employed a more sophisticated FiLM-based integration within SegFormer-B0. Under this rigorous setup the multi-modal models proved competitive, but did not outperform the carefully tuned vision-only baseline (0.545 vs. 0.611 mIoU). Nonetheless, per-class analysis (Fig. 2) reveals benefits for rare classes such as *Crop*, and *Water*, where textual descriptions provide complementary cues.

The underperformance may stem from several factors: a mismatch between the text tokenizer (all-MiniLM-L6-v2) and the captioning models' output style, noisy auto-generated captions, and the need for end-to-end text-encoder fine-tuning. Future work will explore better text-vision alignment, alternative fusion mechanisms, and fine-tuning of the entire multi-modal pipeline to close the gap with the vision-only baseline.

VIII. Resources

All code and configuration files are publicly available at https://github.com/Ksksksksksksksks/di725_project. Training logs, validation curves, per-class IoU breakdowns, and all experiment artifacts are tracked in the Weights & Biases project: https://wandb.ai/e278979-metu-middle-east-technical-university/di725-project/workspace?nw=gj32ledushe.

References

[1] S. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *NeurIPS*, 2021.
[2] H. Liu et al., "LLaVA: Large language and vision assistant," in *arXiv:2304.08485*, 2023.
[3] A. Kirillov et al., "Segment anything," in *ICCV*, 2023.
[4] L. Ma et al., "Deep learning in remote sensing," *IEEE TGRS*, vol. 57, no. 11, pp. 7915–7928, 2019.